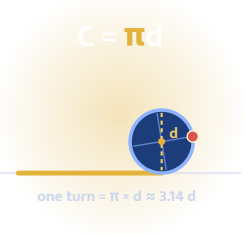
MT Classes, Motihari

Perimeter and Area

Class 9 Maths, Chapter 6: Measuring Space (NCERT Ganita Manjari, Part I)



The story of π, arcs and sectors, Heron's and Brahmagupta's formulas, and the area of a circle, with 54 textbook diagrams, every exercise solved step by step, and a quick quiz.

Start The relay-race puzzle



Fig. 6.1 Athletes at the start of a 4 × 100 m relay race. Why are they not on the same starting line?

In a relay race the runners in the **outer lanes start ahead** of those in the inner lanes, yet everyone finishes at the same line. The gap between the starting points of neighbouring lanes is called the **stagger**.

The big idea

On the curved part of the track, an outer lane is a **bigger circle** than an inner lane, so it is longer. The stagger makes up for this extra length so that **every runner covers exactly the same distance**. To work out the stagger we must know how to find the length around a circle. That is where this chapter begins.

Does the stagger give anyone an unfair advantage?

No. The stagger is chosen so that each runner runs exactly the same distance (for example 400 m). Without it, runners in outer lanes would run more. It only *looks* like they start ahead.

Think and Reflect: My school has a 200 m track instead of a 400 m track. Do we need a smaller stagger for the same 4 × 100 m relay?

Surprisingly, **no, you need a bigger one**. On a bend, the extra distance run by the outer lane is (angle turned) × (difference in radius). It does **not** depend on how big the track is. A 400 m race on a 400 m track turns through 360° once (one lap), but on a 200 m track the same 400 m means **two laps**, so you turn through 720°. With the same lane width, the stagger has to be **twice as large**. (You will be able to check this after Section 6.4.)

6.1 Perimeter of a shape

The **perimeter** of a shape is the total length around its border. Imagine a tiny insect walking once around the edge, never turning back, until it returns to where it started: the distance it walks is the perimeter.

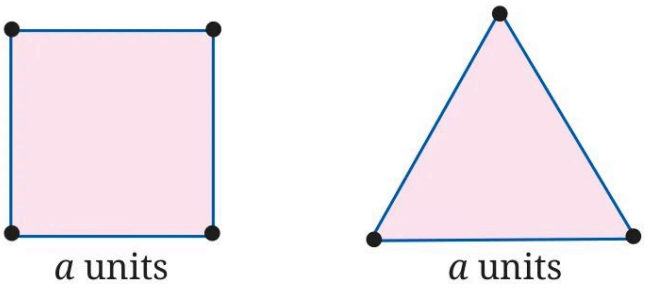


Fig. 6.2 Square: perimeter $4a$ · Equilateral triangle: perimeter $3a$

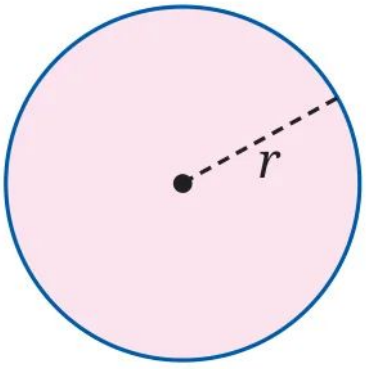


Fig. 6.3 A circle of radius r. What is its perimeter?

Shape	Perimeter
Square of side a	$4a$
Equilateral triangle of side a	$3a$
Rectangle, length a, width b	$2(a + b)$

📝 **Special case**

A square is a rectangle with $a = b$. Put $b = a$ in $2(a + b)$ and you get $4a$. The square formula is a **special case** of the rectangle formula.

Ratio of perimeter to side stays fixed

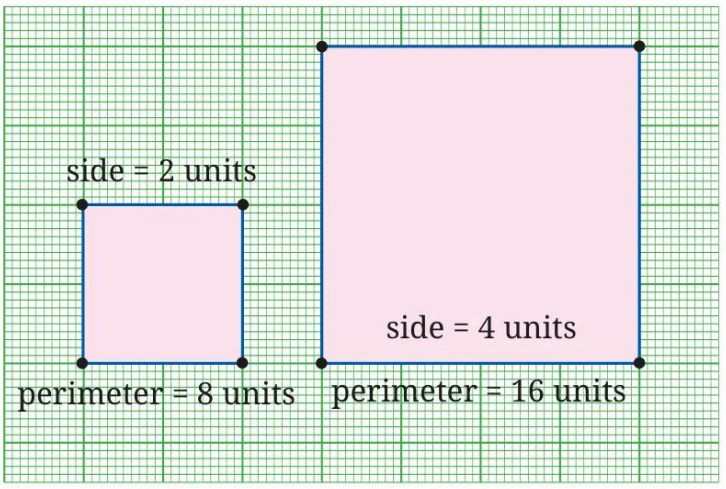


Fig. 6.4 Double the side and the perimeter doubles too: the ratio of perimeter to side is always 4 : 1.

- For **every square**, perimeter : side = **4 : 1**, whatever its size.
- For **every equilateral triangle**, perimeter : side = **3 : 1**.
- Is there a fixed ratio for **circles** too? That is the question of the next section.

– **Think and Reflect: What is the connection between the perimeter of a circle and the 400 m athletics track?**

The curved parts of the track are semicircles. To know how much farther an outer lane is, we need the perimeter (circumference) of circles of different radii.

6.2 The C/D ratio and the story of π

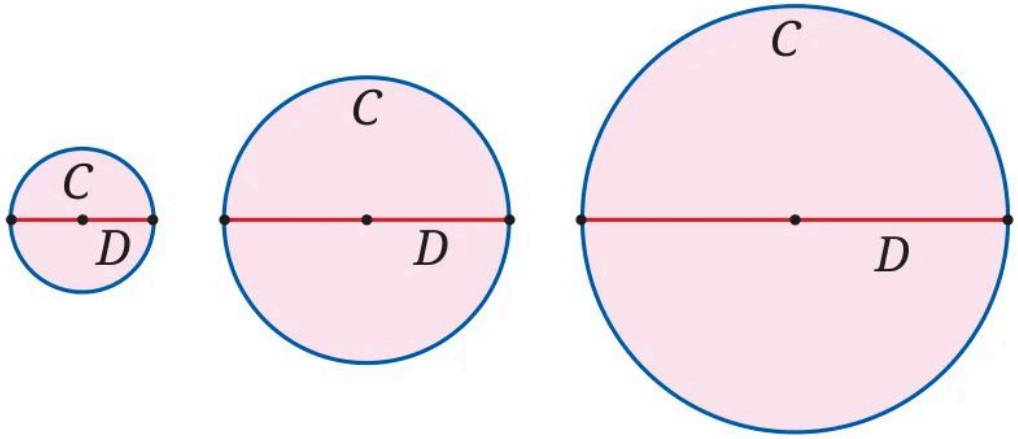


Fig. 6.5 Is circumference ÷ diameter the same for all circles?

The perimeter of a circle is called its **circumference** (*C*). People realised long ago that ***C* ÷ *D* is the same for every circle**, big or small. This fixed number is called **π** (pi, said like "pie").

C ÷ D = π ⟹ C = πD = 2πr

π ≈ 22/7 ≈ 3.14 (π = 3.14159265…). D is the diameter, r the radius, and D = 2r.

🏠 Home measurement

Measure the diameter *D* of a cotton reel. Wrap thin thread tightly around it **20 times**, unwrap it and measure its length *L*. Then **L ÷ (20 × D)** gives π. Winding 20 times makes the small errors in measuring much less important. You should get a value between 3.1 and 3.2.

π's adventurous journey

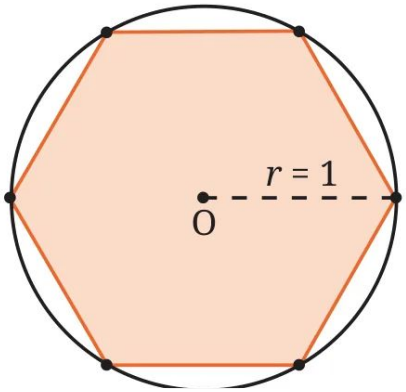


Fig. 6.6 Mesopotamia: the hexagon inside the circle shows π > 3.

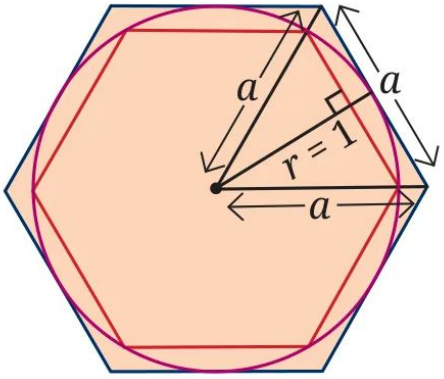


Fig. 6.7 Archimedes trapped π between inscribed and circumscribed polygons.

– **Why does Fig. 6.6 show that π > 3?**

The radius is 1. The hexagon is made of 6 equilateral triangles, so each side is 1 and its perimeter is **6**. The circle goes around the outside of the hexagon, so its circumference 2π × 1 = 2π is **more than 6**. So 2π > 6, i.e. **π > 3**.

– **Why does Fig. 6.7 show that π is between 3 and 2√3?**

The inner hexagon gives π > 3 (as above). For the outer hexagon, the radius 1 is the **height** of each of its 6 equilateral triangles of side a. By Baudhāyana–Pythagoras, $1^2 + (a/2)^2 = a^2$, so $3a^2/4 = 1$ and $a = 2/\sqrt{3}$. Its perimeter is $6a = 12/\sqrt{3} = \mathbf{4\sqrt{3}}$. The circle lies inside it, so $2\pi < 4\sqrt{3}$, i.e. **$\pi < 2\sqrt{3} \approx 3.46$**.

3⅛

Mesopotamia c. 1900 BCE

Hexagon inside a circle ⟹ π > 3. Used π = 3 + 1/8 = **3.125**.

96

Archimedes 250 BCE

Polygons up to **96 sides**: $3\,^{10}/_{71} < \pi < 3\,^{1}/_{7}$.

Pt

Ptolemy 150 CE

377/120 ≈ **3.14167**, for astronomy tables.

Zu

Liu Hui, Zu Chongzhi 263, 480 CE

Polygon with 24,576 sides! Gave **22/7** and **355/113** ≈ 3.1415929, the best value in the world for 800 years.

Ā

Āryabhaṭa 499 CE

62832/20000 = **3.1416**, and called it *āsanna* (approximate): π cannot be an exact fraction.

Br

Brahmagupta 628 CE

Used **√10** ≈ 3.1622: less accurate but easy to use in calculations.

M

Mādhava of Sangamagrāma c. 1400

The **first exact formula** for π, as an infinite series. Got π to 11 decimal places. The beginning of calculus!

π

William Jones, Euler 1706

The symbol π comes from the Greek word *perimetros* (perimeter). Euler made it popular.

π/4 = 1 − 1/3 + 1/5 − 1/7 + ...

Mādhava's series. Later breakthroughs: Nīlakaṇṭha (c. 1500), Machin (1706), Ramanujan (1914), Chudnovsky brothers (1988). Today π is known to hundreds of trillions of digits.

6.3 π is irrational

A **rational** number can be written as a fraction a/b (a, b integers, $b \neq 0$). Its decimal either ends or repeats in a pattern: 1/3 = 0.333..., 1/11 = 0.0909..., 1/7 = 0.142857 142857...

The digits of π go on forever with **no repeating pattern**. π cannot be written as a ratio of two integers, so it is **irrational** (proved by Lambert in 1761). √2 is another irrational number.

Common mistake

π is **not equal** to 22/7. We write **π ≈ 22/7** and **π ≠ 22/7**. In the same way √2 ≈ 1.414 but √2 ≠ 1.414. Since π is irrational there is no "best" fraction for it: for any fraction close to π there is another one that is closer. 355/113 is much better than 22/7.



The digits of π go on forever, like an endless train.

✨ **Fun fact**

"**How I wish I could recollect pi**": count the letters in each word: 3, 1, 4, 1, 5, 9, 2. So π ≈ 3.141592... **Pi Day** is 14 March (3-14) and **Pi Approximation Day** in India is 22 July (22/7).

6.4 Length of an arc

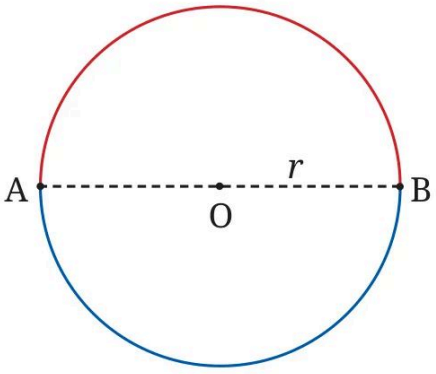


Fig. 6.8 Two equal semicircles: each is 2πr ÷ 2 = **πr**.

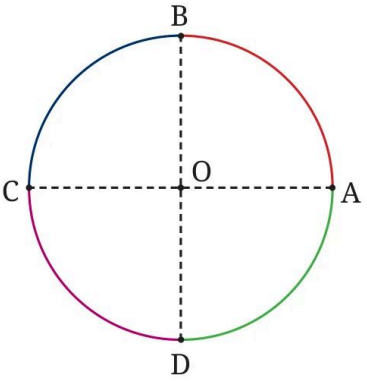


Fig. 6.9 Four equal quarter circles: each is 2πr ÷ 4 = **πr/2**.

Why are the two semicircles equal? **Reflect** the circle in the diameter AB (or turn it through 180°): the red and blue halves swap places exactly. Turning through 90° moves each quarter circle onto the next one, so all four quarters are equal.

- Semicircle = 2πr × **180°/360°**
- Quarter circle = 2πr × **90°/360°**

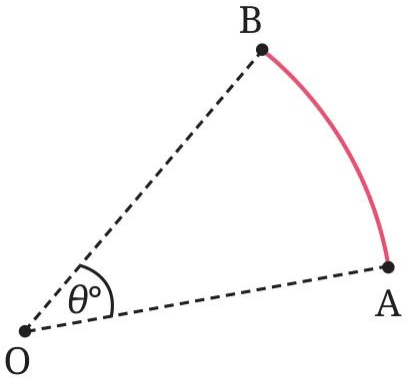


Fig. 6.10 Arc AB subtends angle θ° at the centre O.

Arc length $l = 2\pi r \times \theta/360$

θ = angle the arc subtends at the centre (in degrees). The arc is the fraction $\theta/360$ of the whole circumference.

A closer look at a 400 m athletics track

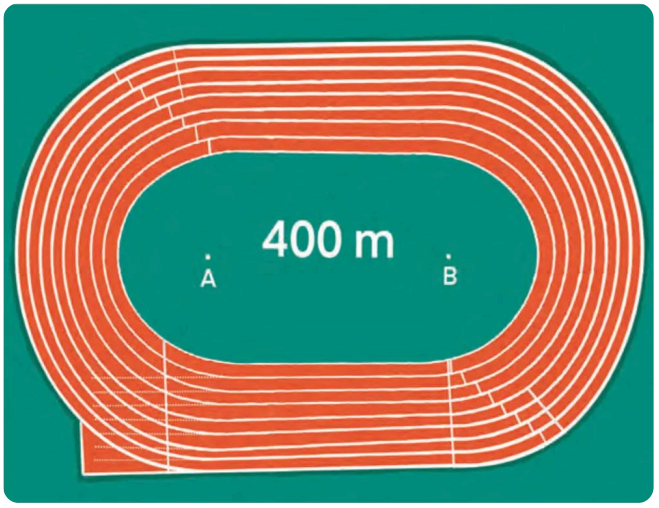


Fig. 6.11 Two straights of 84.39 m and two semicircular bends; innermost radius 36.5 m, each lane 1.22 m wide.

Distance run in lane 1

The athlete runs 0.3 m from the inner border, so the bends have radius 36.5 + 0.3 = **36.8 m**.

Two straights: 2 × 84.39 = 168.78 m. Two semicircles = one full circle: 2 × 3.1416 × 36.8 = 231.22 m.

Total = 168.78 + 231.22 = 400 m

– **Think and Reflect: What is the difference in radius between lanes 1 and 2? What stagger does lane 2 need? Is the stagger between lanes 3 and 2 the same?**

The radius goes up by one lane width, **1.22 m**. In one full lap the two bends make a full circle, so lane 2 runs $2\pi(r + 1.22) - 2\pi r = 2\pi \times 1.22 \approx 2 \times 3.1416 \times 1.22 \approx$ **7.67 m** more. So the lane-2 runner must start about **7.67 m ahead** (for one lap). The extra length depends only on the **difference** in radius (1.22 m), not on the radius itself, so lanes 3 and 2 need the **same** stagger, and so on outwards. (For a 4 × 100 m relay the stagger is shared among the legs run on the bends.)

6.5 Puzzles and paradoxes on perimeter

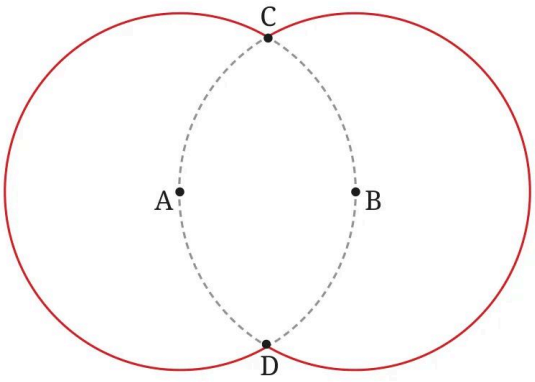


Fig. 6.12 Each circle passes through the other's centre.

Example 1

AB = AC = BC = r, so △ABC is equilateral and ∠CAB = 60°. Likewise ∠BAD = 60°, so **∠CAD = 120°**.

The hidden (dotted) arc of each circle is 120/360 = 1/3 of it, so the outside (red) arc is **2/3** of it.

Perimeter = $2 \times \frac{2}{3} \times 2\pi r = \frac{8}{3}\pi r$

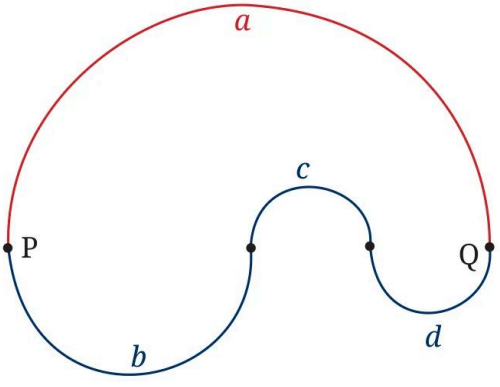


Fig. 6.13 Which path from P to Q is longer?

Example 2: the surprise

Let the radii be a', b', c', d'. A semicircle of radius x has length πx.

Path 1 $= \pi a'$. Path 2 $= \pi(b' + c' + d')$.

But $PQ = 2a' = 2b' + 2c' + 2d'$, so $a' = b' + c' + d'$.

Both paths are equal!

📝 **Remember**

The length of a semicircle is $\pi \times$ radius $= (\pi/2) \times$ diameter. It grows in **proportion** to the diameter, so if you split a diameter into pieces and put a semicircle on each piece, the total length stays the same.

Ex 6.1

Exercise Set 6.1: solutions

Use $\pi = 22/7$ unless told otherwise. Tap a question to see the solution.

1 **The perimeter of a circle is 44 cm. What is its radius?**

$2\pi r = 44 \Rightarrow 2 \times {}^{22}/_{7} \times r = 44 \Rightarrow r = 44 \times 7 \div 44 =$ **7 cm**.

2 **Circumference, correct to 3 significant figures, for radius (i) 7 cm (ii) 10 cm (iii) 12 cm.**

(i) $2 \times {}^{22}/_{7} \times 7 =$ **44.0 cm**
(ii) $2 \times {}^{22}/_{7} \times 10 = 440/7 = 62.857\ldots \approx$ **62.9 cm**
(iii) $2 \times {}^{22}/_{7} \times 12 = 528/7 = 75.428\ldots \approx$ **75.4 cm**

3 **Arc length: (i) r = 3.5 cm, angle 60° (ii) r = 6.3 m, angle 120°.**

(i) $l = 2 \times {}^{22}/_{7} \times 3.5 \times {}^{60}/_{360} = 22 \times {}^{1}/_{6} = 11/3 \approx$ **3.67 cm**
(ii) $l = 2 \times {}^{22}/_{7} \times 6.3 \times {}^{120}/_{360} = 39.6 \times {}^{1}/_{3} =$ **13.2 m**

4 **Perimeter of a sector of radius 14 cm and angle 75°.**

Arc $= 2 \times {}^{22}/_{7} \times 14 \times {}^{75}/_{360} = 88 \times {}^{5}/_{24} = 55/3 \approx 18.33$ cm.
Perimeter = arc + 2 radii = 18.33 + 28 = **46.33 cm** (exactly 139/3 cm).

5 Find the perimeters of the shapes in Fig. 6.14 (i) to (ix).

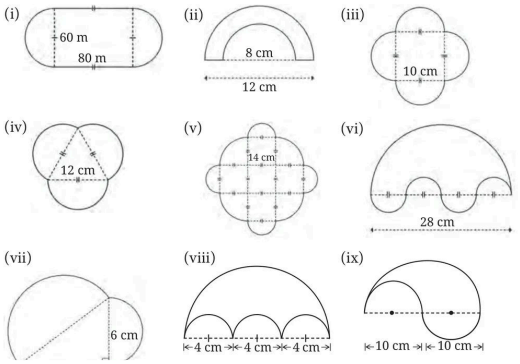


Fig. 6.14 Tap to zoom and follow each part.

(i) Rectangle 80 m × 60 m with a semicircle (diameter 60 m) on each short side. Two semicircles = one circle of diameter 60.
Perimeter = $2 \times 80 + {}^{22}/_{7} \times 60 = 160 + 188.57 =$ **348.57 m**

(ii) Half-ring: outer diameter 12 cm, inner diameter 8 cm, flat ends each $(12 - 8) \div 2 = 2$ cm.
Perimeter = $\pi \times 6 + \pi \times 4 + 2 + 2 = 10\pi + 4 = 220/7 + 4 =$ **35.43 cm**

(iii) Square of side 10 cm with a semicircle on each side (only the arcs are on the border).
Perimeter = $4 \times \pi \times 5 = 20\pi = 440/7 =$ **62.86 cm**

(iv) Equilateral triangle of side 12 cm with a semicircle (diameter 12) on each side.
Perimeter = $3 \times \pi \times 6 = 18\pi = 396/7 =$ **56.57 cm**

(v) All the dashed squares have side 14 cm. The border has 4 quarter circles of radius 14 cm (at the corners) and 4 semicircles of diameter 14 cm (at the ends).
4 quarter circles = one full circle = $2 \times {}^{22}/_{7} \times 14 = 88$ cm. 4 semicircles = $4 \times {}^{22}/_{7} \times 7 = 88$ cm.
Perimeter = 88 + 88 = **176 cm**

(vi) A semicircle on 28 cm, and the base split into 4 equal parts of 7 cm each with a small semicircle on each.
Big arc = ${}^{22}/_{7} \times 14 = 44$ cm. Small arcs = $4 \times {}^{22}/_{7} \times 3.5 = 44$ cm.
Perimeter = **88 cm**

(vii) Right triangle with legs 8 cm and 6 cm, so the hypotenuse is $\sqrt{(64 + 36)} = 10$ cm. A semicircle stands on each side.
Perimeter = $\pi(10 + 8 + 6)/2 = 12\pi = 264/7 =$ **37.71 cm**

(viii) A semicircle on 12 cm and three semicircles on 4 cm.
Perimeter = $\pi \times 6 + 3 \times \pi \times 2 = 12\pi =$ **37.71 cm**

(ix) A semicircle on 20 cm (radius 10), and two semicircles of radius 5 (one up, one down).
Perimeter = $\pi \times 10 + \pi \times 5 + \pi \times 5 = 20\pi =$ **62.86 cm**

📝 **Shortcut**

In (vi), (viii) and (ix) the small semicircles have the same total length as a semicircle on the whole line, just like Example 2.

6 **A car tyre has diameter 56 cm. (i) Distance for one revolution? (ii) Revolutions in 10 km?**

(i) One revolution = circumference = $^{22}/_{7} \times 56$ = **176 cm**.

(ii) 10 km = 10 × 1000 × 100 = 1,000,000 cm. Revolutions = 1,000,000 ÷ 176 = 5681.8..., so about **5682 revolutions** (5681 complete ones).

7 **Total perimeter of all the petals in each flower (Fig. 6.15).**

(i)

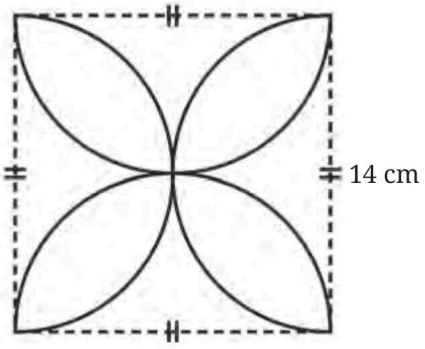


Fig. 6.15A Arcs centred at midpoints of the sides.

(ii)

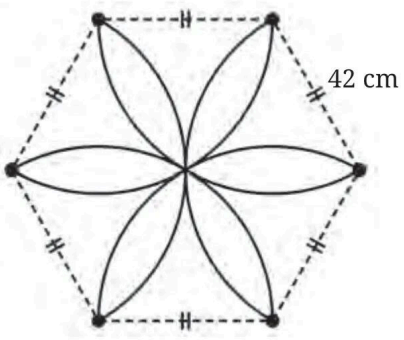


Fig. 6.15B Arcs centred at the vertices.

(i) Each arc is centred at the midpoint of a side, so its radius is 7 cm, and it passes through the two corners and the centre of the square: it is a **semicircle**. The 4 petals are made of 4 such semicircles.

Total = $4 \times {}^{22}/_{7} \times 7$ = **88 cm**

(ii) The centre O and two neighbouring vertices form an **equilateral triangle** of side 42 cm. Each petal edge is an arc of radius 42 cm centred at a vertex, turning through 60°.

One arc = $2 \times {}^{22}/_{7} \times 42 \times {}^{60}/_{360}$ = 44 cm. Each petal has 2 arcs and there are 6 petals: 12 arcs.

Total = 12 × 44 = **528 cm**

8 **The perimeters of two circles are in the ratio 5 : 4. Ratio of their radii?**

$2\pi r_1 : 2\pi r_2 = 5 : 4 \implies r_1 : r_2 =$ **5 : 4**. (Perimeter is proportional to radius.)

6.6–6.7

Rectangle and parallelogram

Area is the amount of space a flat region takes up. The unit is a 1 × 1 square, of area **1 sq. unit** (1 unit^2).

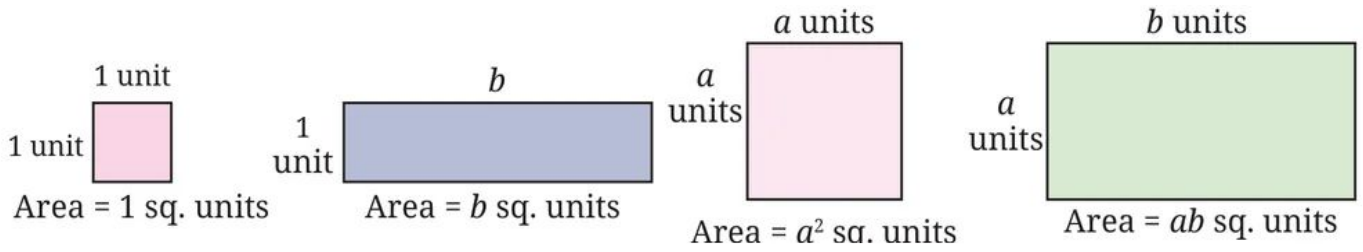


Fig. 6.16 Area of a square and a rectangle.

Rectangle: A = ab · Square: A = a²

Area of a parallelogram

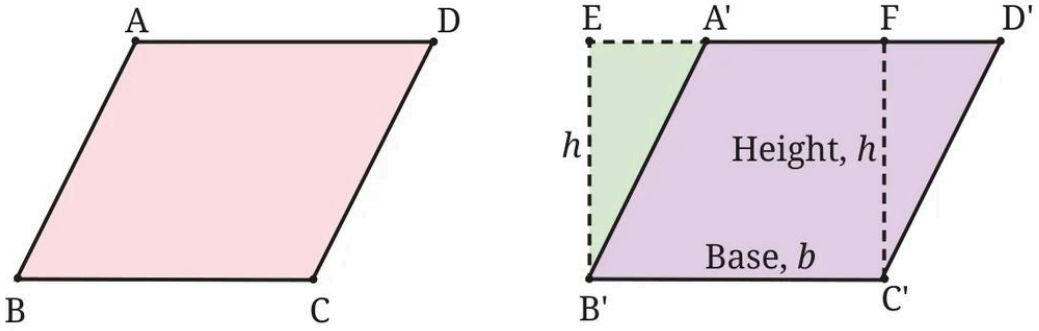


Fig. 6.17 Cut the green triangle off one end and slide it to the other end: the parallelogram becomes a rectangle with the same base and height.

Parallelogram: A = base × height = bh

The height is the perpendicular distance between the parallel sides, not the slanting side.

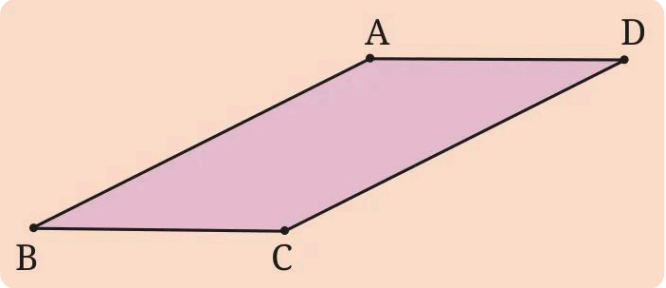


Fig. 6.18 A "thin" parallelogram: the perpendicular from C does not land on AD.

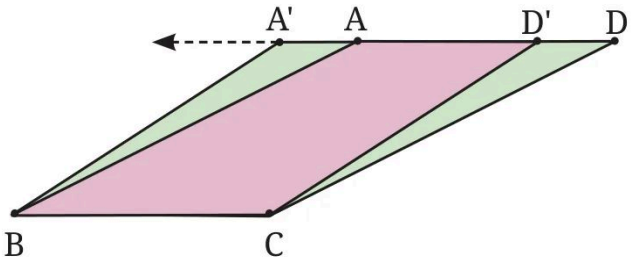


Fig. 6.19 Fix: shift the top side step by step.

Think and Reflect: How do we fix the "gap" for a thin parallelogram?

Pick D′ on DA close to D, and A′ on DA extended with A′A = D′D. $\triangle CDD' \cong \triangle BAA'$, so cutting one off and adding the other keeps the area the same while making the parallelogram A′BCD′ less slanted. Repeat until the foot of the perpendicular lands on the top side; then the cut-and-slide method of Fig. 6.17 works. The base and height never change, so the area is still **bh**.

Think and Reflect: Can we find the area of a parallelogram from its side lengths alone?

No. Keep the sides fixed and change the angle between them: the parallelogram leans over, its height drops and so does its area (it is largest when the angle is 90°, a rectangle, and gets close to 0 as it flattens). Fig. 6.27 shows three rhombuses with sides 3, 3, 3, 3 but areas 9, 8.01 and 5.41.

6.8 Area of a triangle

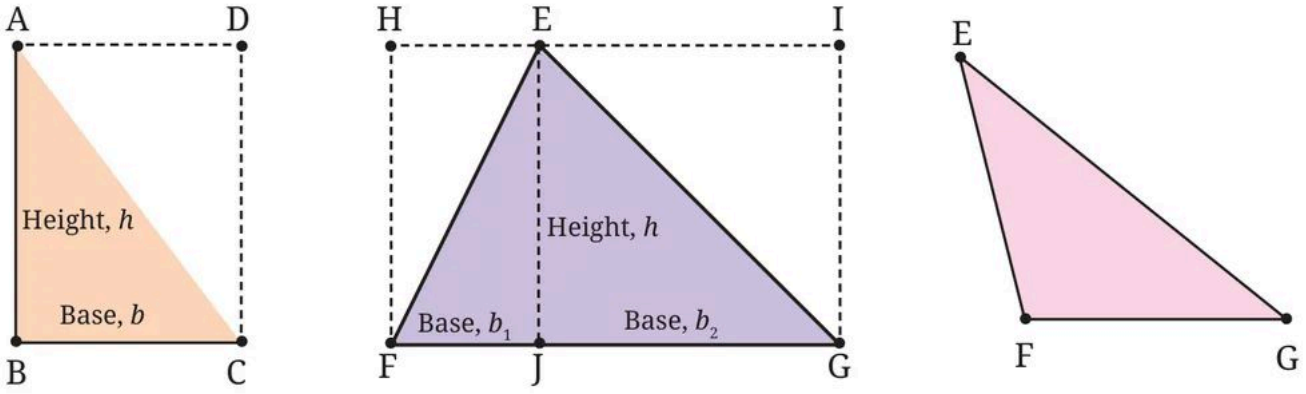


Fig. 6.20A, B Each triangle is exactly half of the rectangle around it.

Enclose the triangle in a rectangle with the same base and height. Each part of the rectangle is cut in half by a side of the triangle, so the triangle is **half the rectangle**.

A nicer way: two copies make a parallelogram

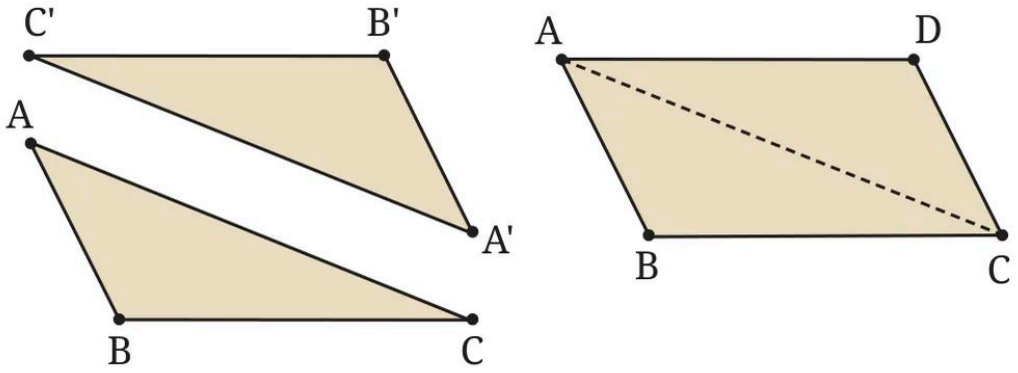


Fig. 6.21 Two congruent triangles fit together to make a parallelogram.

Triangle: A = ½ × base × height = ½bh

– **Why do the two copies make a parallelogram?**

When the copy is turned upside down and placed along BC, the angle ∠B′C′A′ sits next to ∠BCA as alternate angles. Equal alternate angles mean the opposite sides are **parallel**, so both pairs of opposite sides are parallel: a parallelogram.

– **For an obtuse triangle (Fig. 6.20B) the rectangle trick seems to fail. What do we do?**

Draw the height from E to the line FG extended, meeting it at H outside the triangle. Then △EFG = △EHG − △EHF = ½ × HG × h − ½ × HF × h = ½ × (HG − HF) × h = ½ × FG × h. The formula still works.

A property of the median

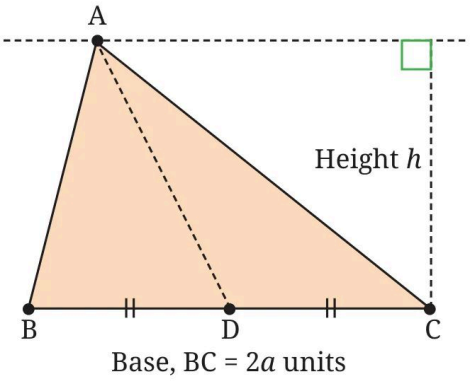


Fig. 6.22 AD is a median: D is the midpoint of BC.

A **median** joins a vertex to the midpoint of the opposite side.

△ABD and △ACD have equal bases (BD = DC = *a*) and the **same height** *h*, so each has area **½ah**.

Theorem

A median of a triangle divides it into two triangles of **equal area**, even though the two triangles usually have different shapes.

– **Think and Reflect: Can △ABD be cut into pieces that rearrange to cover △ACD exactly?**

Yes, it is always possible. In fact any two polygons with equal area can be cut into finitely many pieces and rearranged into each other (the **Wallace–Bolyai–Gerwien theorem**). The book leaves the least number of pieces as a challenge.

– **Think and Reflect: Rectangles with perimeter 40 units. How many? Largest area? Smallest area?**

If the sides are x and $20 - x$ ($0 < x < 20$), there are **infinitely many** such rectangles. Area = $x(20 - x) = 100 - (x - 10)^2$, which is **largest (100 sq. units) when $x = 10$**: the 10 × 10 square. There is **no smallest**: a very thin rectangle (like 19.99 × 0.01) has area as close to 0 as you like, but never 0.

6.8.1 Heron's formula

If you know only the three sides *a*, *b*, *c* of a triangle, you can still find its area. First find the **semi-perimeter** *s*.

$s = \frac{1}{2}(a + b + c)$

$\text{Area} = \sqrt{[\, s(s - a)(s - b)(s - c)\,]}$

Heron of Alexandria. Proved in Grade 10 using Baudhāyana–Pythagoras and $a^2 - b^2 = (a - b)(a + b)$.

Checking it on known triangles

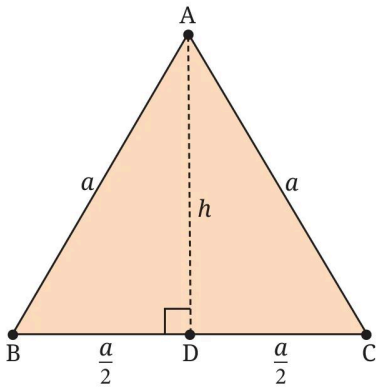


Fig. 6.23 Equilateral, side a

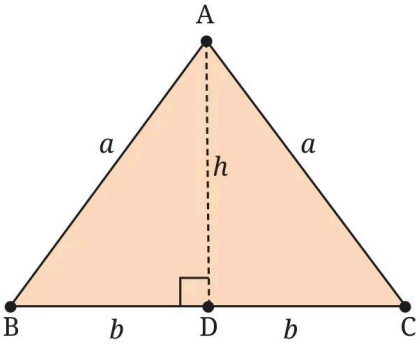


Fig. 6.24 Isosceles, sides a, a, 2b

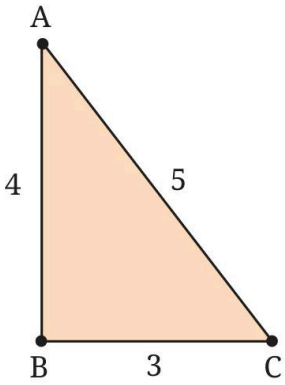


Fig. 6.25 Sides 3, 4, 5

Triangle	Heron's formula	½ × base × height
Equilateral, side *a* (Ex. 3)	s = 3a/2; √(3a/2 · a/2 · a/2 · a/2) = **(√3/4)a²**	h = (√3/2)a; ½ × a × (√3/2)a = **(√3/4)a²**
Isosceles a, a, 2b (Ex. 4)	s = a + b; √((a+b)·b·b·(a−b)) = **b√(a² − b²)**	h = √(a² − b²); ½ × 2b × h = **b√(a² − b²)**
3, 4, 5 (Ex. 5)	s = 6; √(6 × 3 × 2 × 1) = **6**	3² + 4² = 5², right angle at B; ½ × 3 × 4 = **6**

📝 The symbol ∴

∴ means "therefore". You will see it often in maths solutions.

Two more formulas, with circles

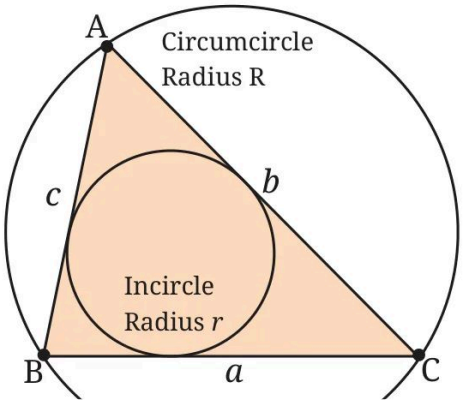


Fig. 6.26 Circumcircle (through the vertices) and incircle (touching the sides).

Area = abc / 4R

R = radius of the **circumcircle**, the one circle through all three vertices.

Area = r(a + b + c) / 2

r = radius of the **incircle**, the one circle touching all three sides.

Cyclic

Brahmagupta's formula

Can we find the area of a 4-gon (quadrilateral) from its four sides alone? **No.** All three shapes below have sides 3, 3, 3, 3, but different areas.

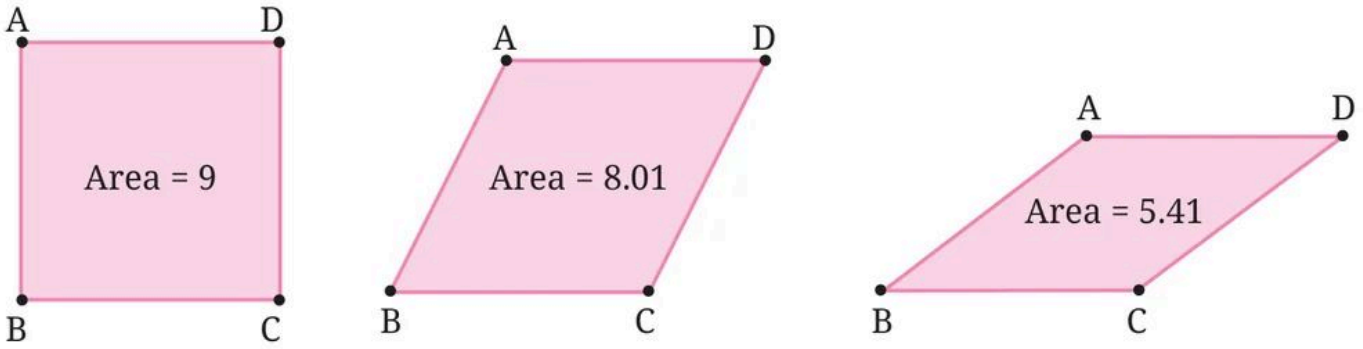


Fig. 6.27 Same sides, different areas. We need extra information, such as an angle, a diagonal, or a special property.

One such special property: the 4-gon is **cyclic**, meaning all four vertices lie on one circle. In 628 CE **Brahmagupta** found a beautiful formula for it.

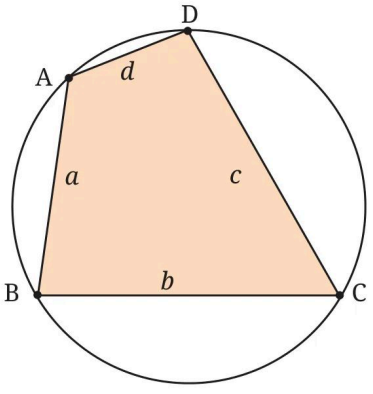


Fig. 6.28 A cyclic 4-gon.

$s = \frac{1}{2}(a + b + c + d)$

$\text{Area} = \sqrt{[(s - a)(s - b)(s - c)(s - d)]}$

Works only for cyclic 4-gons.

Example 6: rectangle (every rectangle is cyclic)

Sides a, b, a, b. $s = a + b$. Area $= \sqrt{(b \cdot a \cdot b \cdot a)} = \sqrt{(ab \cdot ab)} =$ **ab** ✓

Example 7: isosceles trapezium (always cyclic)

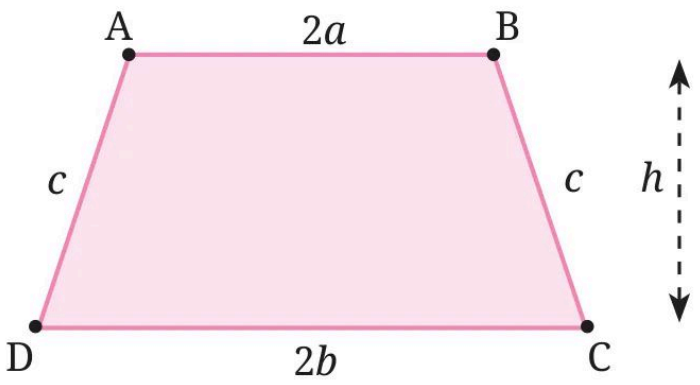


Fig. 6.29 Isosceles trapezium.

Sides 2a, c, 2b, c. $s = a + b + c$. Then $s - 2a = c + b - a$, $s - 2b = c + a - b$, $s - c = a + b$.

Area $= (a + b)\sqrt{[(c + b - a)(c + a - b)]} = (a + b)\sqrt{[c^2 - (b - a)^2]}$.

The height from B is $h = \sqrt{[c^2 - (b - a)^2]}$, so Area $= (a + b)h = \frac{1}{2}(2a + 2b)h$, the usual trapezium formula ✓

✨ Heron is a special case of Brahmagupta

A triangle is a 4-gon with the fourth side **d = 0** (vertices A and D meet), and every triangle is cyclic. Put d = 0: $s = \frac{1}{2}(a + b + c)$ and Area $= \sqrt{[(s - a)(s - b)(s - c)(s - 0)]} = \sqrt{[s(s - a)(s - b)(s - c)]}$, which is **Heron's formula**. Brahmagupta's formula is a **generalisation** of Heron's.

Special case and generalisation

- A square is a special case of a rectangle: $A = a^2$ comes from $A = ab$ with $b = a$.
- Isosceles right triangle from $a^2 + b^2 = c^2$ with $a = b$: $c = a\sqrt{2}$.
- $(a + b)^2 = a^2 + b^2 + 2ab$ is $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ with $c = 0$.

6.9 Squaring a rectangle

To **square** a shape means to construct a square with the same area. Here is how **Baudhāyana** (Śulbasūtra, 800 BCE) squared a rectangle ABCD with AD = a, AB = b, a > b.

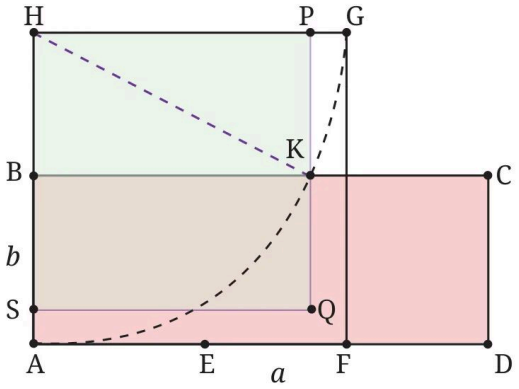


Fig. 6.30 Square HPQS has the same area as rectangle ABCD.

1. Mark E on AD with AE = AB = b.
2. Mark F, the midpoint of ED.
3. Draw square AFGH on AF, with H on AB produced.
4. With centre H, draw arc AG. It cuts BC at K.
5. Through K draw a line parallel to AH, cutting GH at P.
6. Square HPQS on HP has the **same area** as ABCD.

Why it works

AF = (AE + AD)/2 = **(a + b)/2**, so HG = HK = (a + b)/2 (HK is a radius of the arc).

BH = AH − AB = (a + b)/2 − b = **(a − b)/2**.

In right △HKP (KP = BH): $HP^2 = HK^2 - BH^2 = ((a + b)/2)^2 - ((a - b)/2)^2 = (a^2 + 2ab + b^2 - a^2 + 2ab - b^2)/4 =$ **ab**.

So area of square HPQS = HP^2 = ab = area of rectangle ABCD. It is the identity $\mathbf{((a + b)/2)^2 - ((a - b)/2)^2 = ab}$ drawn as a picture.

Think and Reflect: How would you square a triangle?

First turn the triangle into a rectangle of the same area: a triangle of base b and height h has area ½bh, which is a rectangle of base b and height h/2 (cut the triangle at half its height and fold the top pieces down). Then square that rectangle with Baudhāyana's construction.

Ex 6.2 Exercise Set 6.2: solutions

1 Find the area of △ADE in Fig. 6.31.

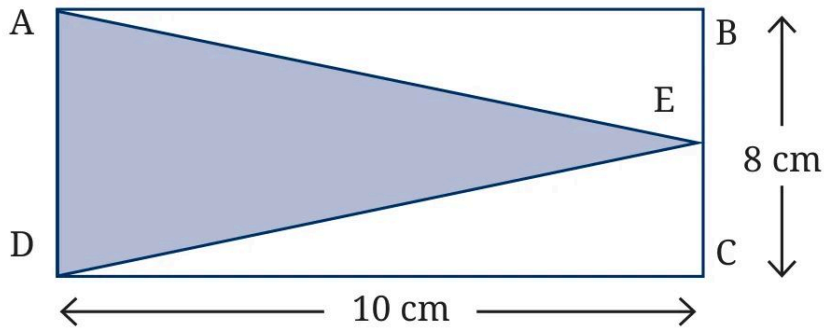


Fig. 6.31

Base AD = 8 cm. E lies on BC, so the height from E to AD is the width of the rectangle, 10 cm.
Area = ½ × 8 × 10 = **40 cm²** (half the rectangle).

2 Trapezium: parallel sides 40 cm and 20 cm, non-parallel sides 26 cm each. Area?

Drop perpendiculars from the ends of the 20 cm side. They cut off (40 − 20) ÷ 2 = 10 cm on each side of the 40 cm side.
Height h = $\sqrt{(26^2 - 10^2)} = \sqrt{(676 - 100)} = \sqrt{576} = 24$ cm.
Area = ½ × (40 + 20) × 24 = **720 cm²**.

3 Sides 8 cm and 11 cm, perimeter 32 cm. Area?

Third side = 32 − 8 − 11 = 13 cm. s = 16.
Area = $\sqrt{(16 \times 8 \times 5 \times 3)} = \sqrt{1920} = 8\sqrt{30} \approx 43.82$ **cm²**.

4 Sides in the ratio 3 : 5 : 7, perimeter 300 m. Area?

Sides 3k + 5k + 7k = 15k = 300, k = 20: sides 60 m, 100 m, 140 m. s = 150.
Area = $\sqrt{(150 \times 90 \times 50 \times 10)} = \sqrt{6{,}750{,}000} = 1500\sqrt{3} \approx 2598$ **m²**.

5 **Rhombus with one diagonal twice the other, area 128 cm^2. Shorter diagonal?**

Area of a rhombus = ½ × d_1 × d_2. With d and 2d: ½ × d × 2d = d^2 = 128.

d = √128 = **8√2 ≈ 11.31 cm**.

6 **ABCD is a parallelogram; P and Q are on AB. Ratio area(△PCD) : area(△QCD)?**

Both triangles have the same base CD, and P, Q lie on AB, which is parallel to CD. So both have the same height (the distance between AB and CD). Ratio = **1 : 1**. Each is half the parallelogram.

7 **O is any point on diagonal PR of parallelogram PQRS. Prove area(△PSO) = area(△PQO).**

Let the diagonals meet at M. Diagonals of a parallelogram bisect each other, so M is the midpoint of QS.

In △PQS, PM is a median, so area(△PQM) = area(△PSM). In △OQS, OM is a median, so area(△OQM) = area(△OSM).

Subtract (or add, if O is beyond M): area(△PQO) = area(△PSO). ∎

Another way: △PSO and △PQO share the base PO, and S and Q are at equal distances from the line PR (as QM = MS).

8 **Joining the midpoints of the sides of a 4-gon in order gives a parallelogram of half the area. Prove it.**

Let ABCD have midpoints P (AB), Q (BC), R (CD), S (DA).

In △ABD, BS is a median, so area(ABS) = ½ area(ABD). In △ABS, SP is a median, so area(APS) = ½ area(ABS) = **¼ area(ABD)**. Similarly area(CQR) = ¼ area(CBD).

Adding: area(APS) + area(CQR) = ¼ area(ABCD). In the same way area(BPQ) + area(DRS) = ¼ area(ABCD).

The four corner triangles take ½ of ABCD, so PQRS = **½ area(ABCD)**. ∎

– **9** **In △ABC, D is the midpoint of BC and P is any point on median AD. Show area(△ABP) = area(△ACP).**

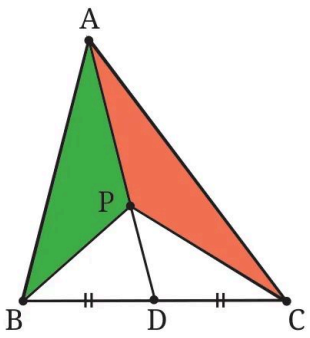


Fig. 6.32

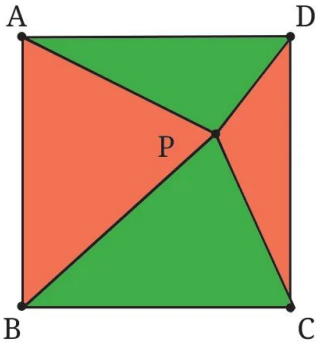


Fig. 6.33

AD is a median of △ABC: area(ABD) = area(ACD).

PD is a median of △PBC: area(PBD) = area(PCD).

Subtract: area(ABD) − area(PBD) = area(ACD) − area(PCD), i.e. **area(△ABP) = area(△ACP)**. ∎

– **10** **Square ABCD with P inside. Ratio of red (△PAB + △PCD) to green (△PBC + △PDA)?**

Let the side be s. △PAB and △PCD have bases AB = CD = s, and their heights add up to the distance between AB and CD, which is s. So red = ½s × (h_1 + h_2) = ½s^2. Green is the rest, also ½s^2. Ratio = **1 : 1**, wherever P is.

– **11** **D is the midpoint of AB, P is on BC, Q is on AB with CQ ∥ PD. Prove area(△BPQ) = ½ area(△ABC).**

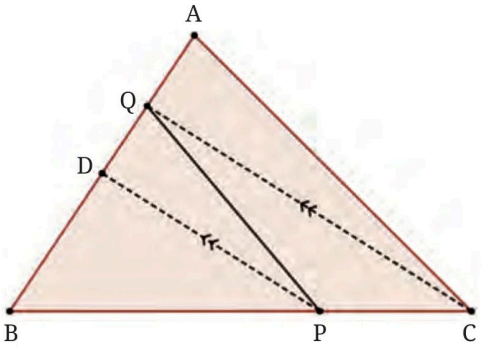


Fig. 6.34

△PDQ and △PDC have the same base PD and lie between the same parallels PD and CQ, so they have **equal area**.

area(BPQ) = area(BPD) + area(PDQ) = area(BPD) + area(PDC) = area(BDC).

CD is a median of △ABC, so area(BDC) = ½ area(ABC). Hence **area(△BPQ) = ½ area(△ABC)**. ∎

6.10 Area of a circle

For every square, perimeter^2 : area = $(4a)^2 : a^2$ = **16 : 1**. For every equilateral triangle it is $9a^2 : (\sqrt{3}/4)a^2$ = **36 : √3**. For any shape this ratio depends only on the **shape**, not its size. So for circles too, **C^2 : A** is a fixed number, and so is **A : r^2**.

Who	Formula for the area of a circle
Babylonians (before 1500 BCE)	$A \approx C^2/12$
Egyptians (1500 BCE), Baudhāyana Śulbasūtra (800 BCE)	$A \approx (8d/9)^2 = (256/81)r^2 \approx 3.16r^2$
Archimedes (c. 250 BCE)	**$A = \pi r^2$** exactly, the same π as in $C = 2\pi r$

Archimedes: "The area of a circle equals the area of a right triangle whose two legs are the radius and the circumference."

Area of circle = ½ × circumference × radius = ½ × 2πr × r = πr²

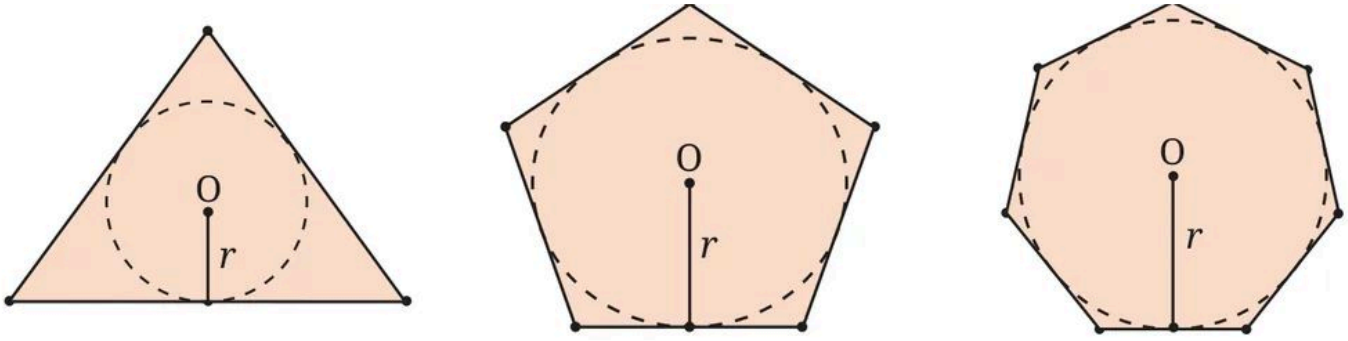


Fig. 6.36 Area of a regular polygon = ½ × perimeter × radius of its incircle. With more and more sides, the polygon becomes the circle.

Nīlakaṇṭha's picture proof (c. 1500)

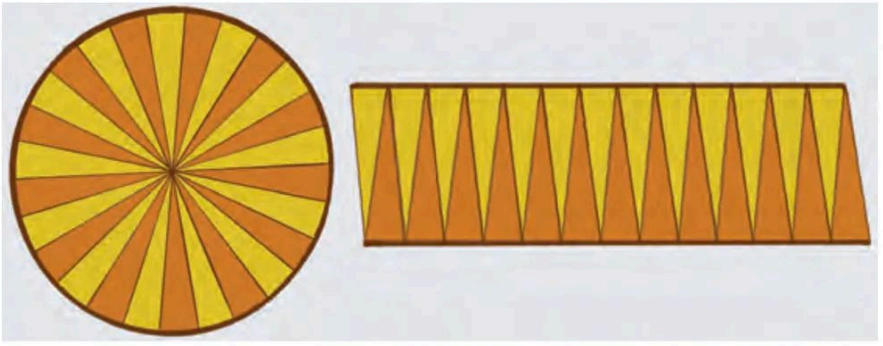


Fig. 6.37 Cut the circle into thin slices and arrange them top-to-tail.

As the slices get thinner, the shape becomes a **parallelogram** with

- base = half the circumference = **πr** (half the slices' arcs are on top, half on the bottom),
- height = **r**.

Area of circle = πr × r = πr²

– **Think and Reflect: Why do people love circular shapes?**

Practical reasons: wheels roll smoothly; a circle encloses the most area for a given length of boundary (wall, fence or rope); round pots and towers are strong and easy to make with a peg and rope. Other reasons: circles look beautiful and balanced and have been symbols of the sun, time and unity in many cultures.

6.10.1 Sector and segment

A **sector** is the region between an arc and the two radii to its ends (a slice of pizza). A **segment** is the region between an arc and its chord.

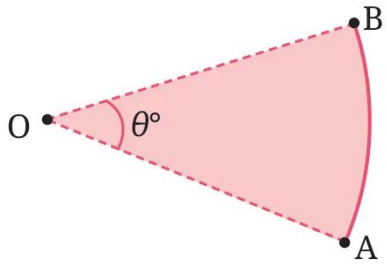


Fig. 6.38 Sector, angle θ°

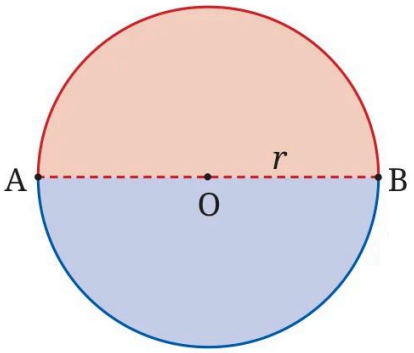


Fig. 6.39 Semicircular disc = ½πr²

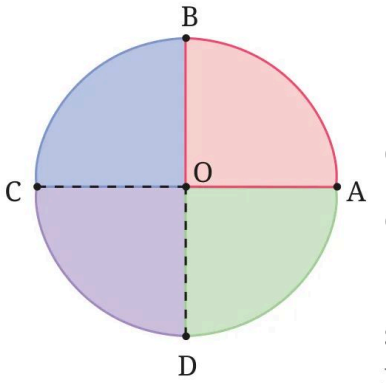


Fig. 6.40 Quarter disc = ¼πr²

Area of sector = πr² × θ/360

Compare: arc length = 2πr × θ/360. Both use the same fraction θ/360 of the whole circle.

Area of segment = area of sector − area of triangle

📝 **Minor and major**

The smaller sector/segment is the **minor** one, the larger is the **major** one. Major = whole circle − minor.

Ex 6.3

Exercise Set 6.3: solutions

1 **Area of a sector of radius 7 cm, angle 60°.**

$\frac{22}{7} \times 7 \times 7 \times \frac{60}{360} = 154 \times \frac{1}{6} = 77/3 \approx$ **25.67 cm²**

2 **Area of a quadrant of a circle with circumference 44 cm.**

$2\pi r = 44 \Rightarrow r = 7$ cm. Circle area $= \frac{22}{7} \times 49 = 154$ cm². Quadrant $= 154 \div 4 =$ **38.5 cm²**.

3 **Minute hand 7 cm long. Area swept in 10 minutes.**

In 60 minutes it turns 360°, so in 10 minutes it turns 60°.

Area $= \frac{22}{7} \times 49 \times \frac{60}{360} = 154/6 \approx$ **25.67 cm²**.

4 **Chord of a circle of radius 10 cm subtends 90°. Areas of (i) minor sector (ii) major sector (π ≈ 3.14).**

Circle area = 3.14 × 100 = 314 cm^2.
(i) Minor sector = 314 × 90/360 = **78.5 cm^2**
(ii) Major sector = 314 × 270/360 = **235.5 cm^2**

5 **Chord of a circle of radius 15 cm subtends 60°. Minor and major segments (π ≈ 3.14, √3 ≈ 1.73).**

Sector = 3.14 × 225 × 60/360 = 117.75 cm^2.
The triangle formed by the chord and two radii is **equilateral** (two sides 15 cm and angle 60°), so its area = (√3/4) × 15^2 = 1.73 × 225 ÷ 4 = 97.31 cm^2.
Minor segment = 117.75 − 97.31 = **20.44 cm^2** (more exactly 20.4375).
Circle = 3.14 × 225 = 706.5 cm^2. Major segment = 706.5 − 20.44 = **686.06 cm^2**.

6 **Two wipers, each 28 cm long, sweep 120° without overlapping. Total area cleaned per sweep.**

One wiper = $^{22}/_{7}$ × 28 × 28 × $^{120}/_{360}$ = 2464 ÷ 3 = 821.33 cm^2.
Two wipers = 4928/3 ≈ **1642.67 cm^2**.

7* **A chord of a circle of radius r subtends 60°. Show the minor segment = r^2(π/6 − √3/4).**

Sector = πr^2 × 60/360 = πr^2/6. The triangle is equilateral with side r, area (√3/4)r^2.
Minor segment = πr^2/6 − (√3/4)r^2 = **r^2(π/6 − √3/4)** ≈ 0.0906r^2. ∎
Note: the textbook prints this as πr^2(1/6 − √3/4), which would be negative. The π belongs only with 1/6, as shown here.

8* **Equilateral triangle inscribed in a circle of radius r. Show area of triangle : area of circle = 3√3 / 4π ≈ 0.413.**

Join the centre O to the vertices: three triangles, each with two sides r and angle 120° between them. The side of the big triangle is then r√3 (the perpendicular from O to a side is r/2, so half a side is √(r^2 − r^2/4) = (√3/2)r).
Area of triangle = (√3/4)(r√3)^2 = (3√3/4)r^2.
Ratio = (3√3/4)r^2 ÷ πr^2 = **3√3/(4π) ≈ 0.413**. ∎

– **9* Square inscribed in a circle of radius r. Show the ratio of areas = 2/π ≈ 0.637.**

The diagonal of the square is the diameter 2r. Area of a square = ½ × diagonal^2 = ½ × 4r^2 = 2r^2. Ratio = 2r^2 ÷ πr^2 = **2/π ≈ 0.637**. ∎

– **10* Regular hexagon inscribed in a circle of radius r. Show the ratio = 3√3/2π ≈ 0.827. Why is it twice the answer to Q8?**

The hexagon is 6 equilateral triangles of side r: area = 6 × (√3/4)r^2 = (3√3/2)r^2. Ratio = **3√3/(2π) ≈ 0.827**.

Why twice? Join every other vertex of the hexagon to get the inscribed equilateral triangle of Q8. It covers the 3 triangles OAC, OCE, OEA, each equal in area to one of the 6 small triangles. So the triangle is 3 parts and the hexagon is 6 parts: the hexagon is exactly twice the triangle.

End

End-of-chapter exercises: solutions

Use π = 22/7 unless told otherwise. Questions marked * are challenge problems.

1 Draw area figures for $(a + b)(a - b) = a^2 - b^2$ and $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.

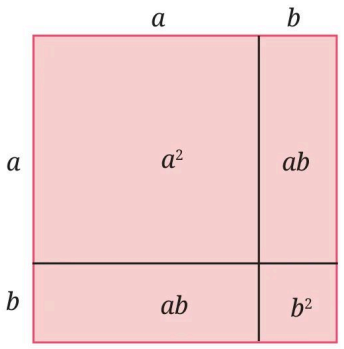
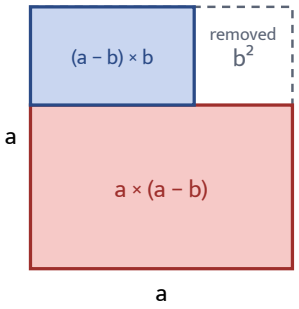
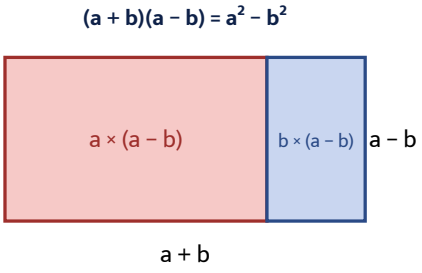


Fig. 6.41 $(a + b)^2 = a^2 + 2ab + b^2$

$(a + b)(a - b) = a^2 - b^2$: from a square of side a, cut away a square of side b from a corner. The leftover L-shape (area $a^2 - b^2$) splits into two strips. Turn the small strip and put it at the end of the big one: you get a rectangle $(a + b)$ long and $(a - b)$ wide.

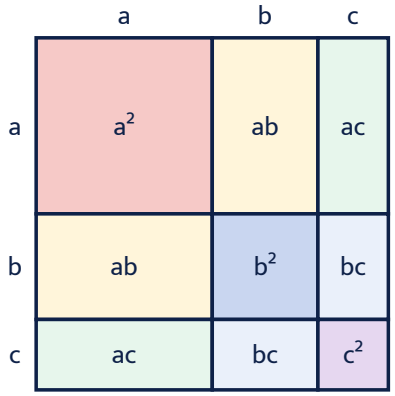


Square a × a minus the corner b × b



Rearranged into a rectangle

$(a + b + c)^2$: split each side of a square of side $(a + b + c)$ into a, b, c. You get 9 pieces: a^2, b^2, c^2 on the diagonal, and two each of ab, bc and ca.



$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

2 Isosceles triangle, perimeter 40 cm, equal sides 15 cm each. Area?

Base = 40 − 30 = 10 cm. Height = $\sqrt{(15^2 - 5^2)} = \sqrt{200} = 10\sqrt{2}$ cm.

Area = ½ × 10 × $10\sqrt{2}$ = **$50\sqrt{2} \approx 70.71$ cm²**. (Heron: s = 20, $\sqrt{(20 \times 5 \times 5 \times 10)} = \sqrt{5000} = 50\sqrt{2}$ ✓)

3 Isosceles triangle with base 10 cm and area 60 cm². Equal sides?

½ × 10 × h = 60 ⟹ h = 12 cm. The height bisects the base: equal side = $\sqrt{(5^2 + 12^2)} = \sqrt{169}$ = **13 cm**.

4 Right triangle of area 54 cm² with one leg 12 cm. Perimeter?

½ × 12 × other leg = 54 ⟹ other leg = 9 cm. Hypotenuse = $\sqrt{(144 + 81)} = \sqrt{225}$ = 15 cm.

Perimeter = 12 + 9 + 15 = **36 cm**.

5 Sides in the ratio 2 : 3 : 4, perimeter 45 cm. Area?

9k = 45, k = 5: sides 10, 15, 20 cm. s = 22.5.

Area = $\sqrt{(22.5 \times 12.5 \times 7.5 \times 2.5)} = \sqrt{5273.4375}$ = **$(75\sqrt{15})/4 \approx 72.62$ cm²**.

6 Sides 7 cm, 24 cm, 25 cm. Area in two ways.

Way 1: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$, so the triangle is right-angled. Area = ½ × 7 × 24 = **84 cm²**.

Way 2 (Heron): s = 28. Area = $\sqrt{(28 \times 21 \times 4 \times 3)} = \sqrt{7056}$ = **84 cm²**.

7 Bicycle wheel of diameter 60 cm rotates 100 times. Distance?

One turn = $^{22}/_{7}$ × 60 = 1320/7 cm. 100 turns = 132000/7 ≈ 18,857 cm ≈ **188.57 m**.

8 **Area of a quadrant of a circle with circumference 66 cm.**

$2 \times {}^{22}/_{7} \times r = 66 \implies r = 10.5$ cm. Circle = ${}^{22}/_{7} \times 10.5^2 = 346.5$ cm^2. Quadrant = **86.625 cm^2**.

9 **Car wheel of radius 28 cm. Distance in one turn, and turns in 1 km?**

One turn = $2 \times {}^{22}/_{7} \times 28 =$ **176 cm**. 1 km = 100,000 cm, so turns = 100,000 ÷ 176 ≈ **568.2** (about 568 turns).

10* **Two rectangles have the same area and the same perimeter. Are they congruent?**

Yes. Let the sides be x and y with $x + y = p$ (half the perimeter) and $xy = A$. Then x and y are the two roots of $t^2 - pt + A = 0$, so they are fixed by p and A. Any other rectangle with the same p and A has the same pair of sides, so it is congruent. (This is not true for other shapes: two different triangles can have equal area and perimeter.)

11 **Using the figure, show the area of a trapezium is ½(a + b)h.**

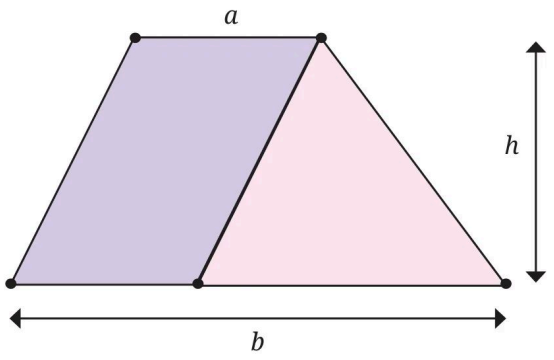


Fig. 6.42

The trapezium is a parallelogram (base a, height h) plus a triangle (base b − a, height h).
Area = ah + ½(b − a)h = ½(2a + b − a)h = **½(a + b)h**. ∎

12 **Divide a trapezium into two triangles to get the same formula.**

Draw a diagonal. One triangle has base a (top side) and the other has base b (bottom side); both have height h, the distance between the parallel sides.
Area = ½ah + ½bh = **½(a + b)h**. ∎

13 Use two identical trapeziums to make a parallelogram.

Turn one copy upside down (rotate it 180°) and join it to the first along a slanting side. The top becomes a + b and the bottom b + a, so the shape is a parallelogram with base (a + b) and height h. Its area is (a + b)h, so one trapezium = **½(a + b)h**.

14 Show that the area of a kite is half the product of its diagonals, (i) using algebra, (ii) using geometry.

In kite ABCD (AB = AD, CB = CD) the diagonals are perpendicular: BD ⊥ AC, meeting at O.

(i) Algebra: Area = area(ABC) + area(ADC) = ½ × AC × OB + ½ × AC × OD = ½ × AC × (OB + OD) = **½ × AC × BD**.

(ii) Geometry: Draw the rectangle with sides AC and BD around the kite (sides parallel to the diagonals). The diagonals cut it into 4 small rectangles, and each side of the kite cuts one of them exactly in half. So the kite is half the rectangle: **½ × d_1 × d_2**.

15 Fitting congruent shapes: sides doubled or tripled.

(i) PQRS has area 2a × 2b = 4ab = 4 × area(ABCD). **Yes**, 4 copies fit in a 2 × 2 arrangement.

(ii) Doubling every side doubles the base and the height, so the area becomes 2 × 2 = **4 times**.

Yes: join the midpoints of the sides of △PQR; this makes 4 triangles, each with sides a, b, c (midpoint theorem), so each is congruent to △ABC.

(iii) Tripling gives 3 × 3 = **9 times** the area. **Yes**: divide each side of △PQR into 3 equal parts and draw lines parallel to the sides. You get 9 small triangles (6 pointing one way and 3 the other), all congruent to △ABC.

– **16*** **What fraction of the triangle (Fig. 6.43) and of the square (Fig. 6.44) is shaded?**

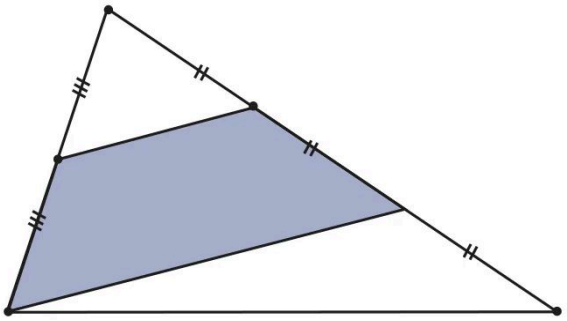


Fig. 6.43

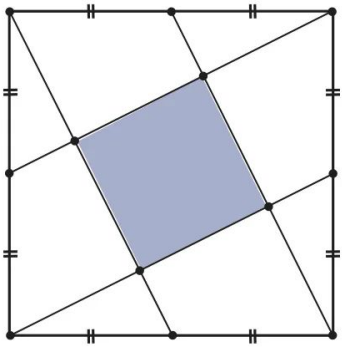


Fig. 6.44

Fig. 6.43: Call the top vertex T. One side is cut at its midpoint M, the other side into three equal parts at P_1, P_2 (P_1 nearer T). The shaded part = △(T, bottom-left vertex, P_2) − △TMP_1.
△ with T and P_2 = $^2/_3$ of the whole (same height, base 2/3). △TMP_1 = $^1/_2 \times {}^1/_3 = {}^1/_6$ of the whole.
Shaded = 2/3 − 1/6 = **1/2**.

Fig. 6.44: Each corner is joined to the midpoint of a side, making four slanted lines and four small right triangles next to the shaded square. Turn each small triangle half a turn about the midpoint it touches: it fills the notch on the other side of that midpoint. The big square is then rebuilt as **5 copies** of the shaded square (a plus shape). So the shaded square is **1/5** of the big square. Check: with side 2, the shaded square has side 2/√5, so area 4/5 out of 4.

– **17** **What fraction of the rectangle is covered by the circles (Figs. 6.45, 6.46)?**

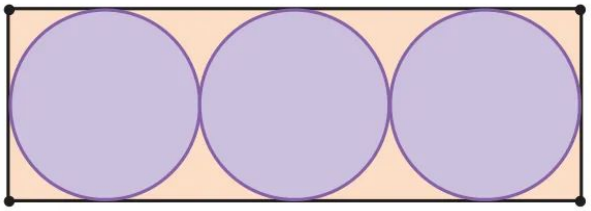


Fig. 6.45

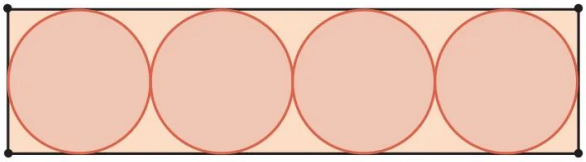


Fig. 6.46

With n circles of radius r: rectangle = $2nr \times 2r = 4nr^2$. Circles = $n\pi r^2$.
Fraction = $n\pi r^2 \div 4nr^2$ = **$\pi/4 \approx 0.785$** for both.

18 **Make and prove a conjecture for n circles (10, 20, 50).**

Conjecture: however many circles there are in a row, they always cover **π/4 (about 78.5%)** of the rectangle.

Proof: as in Q17, $n\pi r^2 \div (2nr \times 2r) = \pi/4$; the n cancels. For 10, 20 or 50 circles the answer is the same.

19* **Nine identical rectangles make a large rectangle of area 72 cm^2. Perimeter of each small rectangle?**

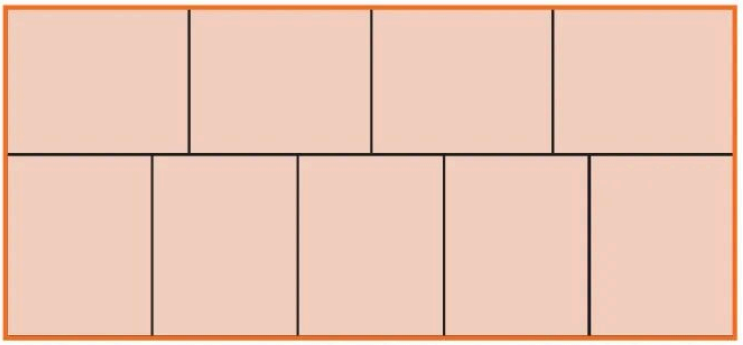


Fig. 6.47

Let each small rectangle be L × W (L longer). The top row has 4 lying flat (width 4L), the bottom row 5 standing up (width 5W). So **4L = 5W**.

Each has area 72 ÷ 9 = 8 cm^2, so LW = 8. With W = 4L/5: $4L^2/5 = 8$, $L^2 = 10$, **L = √10 ≈ 3.16 cm**, W = 4√10/5 ≈ 2.53 cm.

Perimeter = 2(L + W) = 2 × 9√10/5 = **18√10/5 ≈ 11.38 cm**.

– **20*** **Lines from a vertex to the trisection points of the opposite side: show the blue and red triangles are equal.**

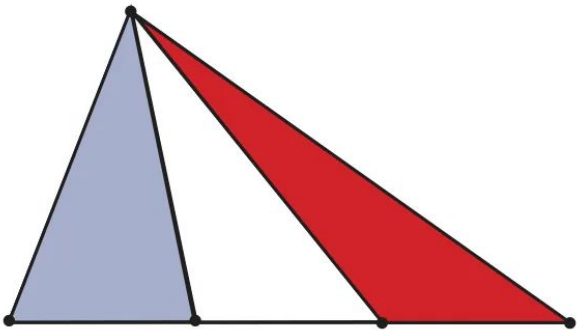


Fig. 6.48

The blue and red triangles have the same top vertex, so the same height, and equal bases (each one-third of the base). So their areas are **equal**.

Cutting idea: cut each triangle along the line joining the midpoints of its two slanting sides, and give the small top piece a half turn about one of those midpoints. Each triangle becomes a **parallelogram** with base ⅓ of the big base and half the height. Two parallelograms with the same base and height can be cut and slid into each other (as in Figs. 6.17 and 6.19). So the blue pieces can be rearranged to cover the red triangle.

– **21*** **Quarter circle and two semicircles in a square: show regions A and B have equal area.**

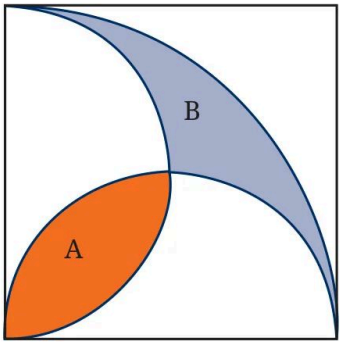


Fig. 6.49

Let the side be 2r. Quarter circle area = $\frac{1}{4}\pi(2r)^2 = \pi r^2$. Each semicircle has area $\frac{1}{2}\pi r^2$, so the two together have area πr^2, the same as the quarter circle.

The two semicircles overlap in A. So the region they cover = $\pi r^2 - A$. The quarter circle = that region + B = $\pi r^2 - A + B$.

So $\pi r^2 = \pi r^2 - A + B$, giving **A = B**. ∎

22* **Four semicircles inside a square of side 2: perimeter and area of the 4-petal flower.**

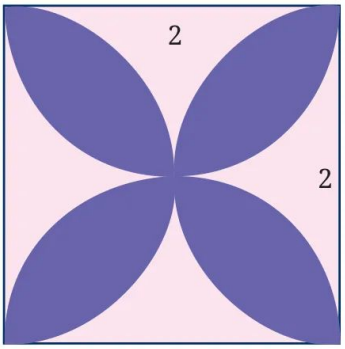


Fig. 6.50

Each semicircle has radius 1. Each petal is where two neighbouring semicircles overlap, and its two edges are **quarter circles** of radius 1.

Perimeter: 8 quarter arcs = 2 full circles of radius 1 = $4\pi = 88/7 \approx$ **12.57 units**.

Area: one petal = 2 × (quarter circle − right triangle) = $2(\pi/4 - \tfrac{1}{2}) = \pi/2 - 1$. Four petals = **$2\pi - 4 \approx$ 2.29 sq. units**.

23* **Concentric circles; chord BC = l of the larger touches the smaller. Show the ring area is $\tfrac{1}{4}\pi l^2$.**

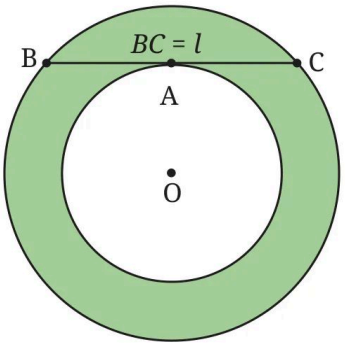


Fig. 6.51

OA ⊥ BC (radius ⊥ tangent), and A is the midpoint of BC, so AB = l/2. In right △OAB: $R^2 - r^2 = OB^2 - OA^2 = (l/2)^2$.

Ring area = $\pi R^2 - \pi r^2 = \pi(R^2 - r^2) =$ **$\tfrac{1}{4}\pi l^2$**. ∎ (You do not need to know R or r!)

24* Semicircles on the sides of a right triangle: show Area(A) + Area(B) = Area(C).

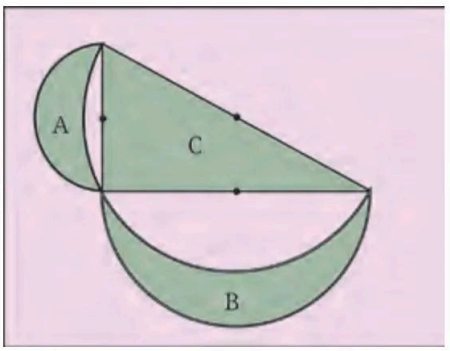


Fig. 6.52

A semicircle on a side of length x has area $\tfrac{1}{2}\pi(x/2)^2 = (\pi/8)x^2$. With legs a, b and hypotenuse c: $A + B = (\pi/8)(a^2 + b^2) = (\pi/8)c^2 = \mathbf{C}$, because $a^2 + b^2 = c^2$. ∎

25* Two circles of radius r through each other's centres: area of the region common to both.

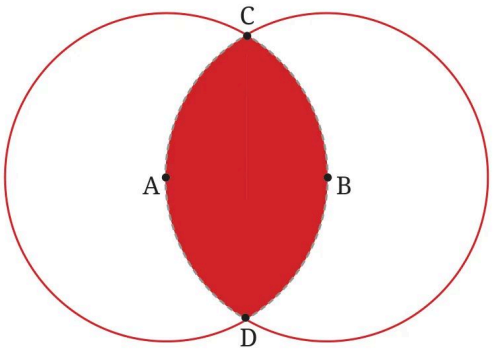


Fig. 6.53

As in Example 1, ∠CAD = ∠CBD = 120°. The lens is two segments, each = sector 120° − △ with two sides r and angle 120°.

One segment = $\pi r^2/3 - (\sqrt{3}/4)r^2$. Lens = 2 × that = $\mathbf{r^2(2\pi/3 - \sqrt{3}/2) \approx 1.23r^2}$.

26* Three triangles A, B, C in a rectangle: show the rectangle's area is 2(A + C)(B + C)/C.

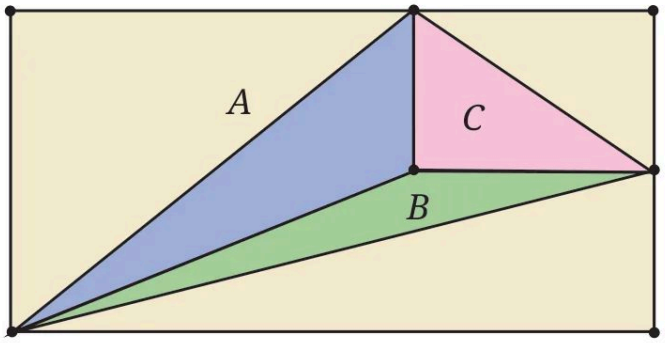


Fig. 6.54

Let the rectangle be w × h with the bottom-left corner at O. The top point is at distance q from the left side, the right point at height r. Triangle C is right-angled with legs (h − r) and (w − q).

A = ½q(h − r), B = ½r(w − q), C = ½(h − r)(w − q).

A + C = ½(h − r)w and B + C = ½(w − q)h.

(A + C)(B + C) = ¼wh(h − r)(w − q) = ¼wh × 2C = ½wh·C.

So **wh = 2(A + C)(B + C)/C.** ∎

27* Quarter circle, semicircle and triangle: show the two shaded regions are equal.

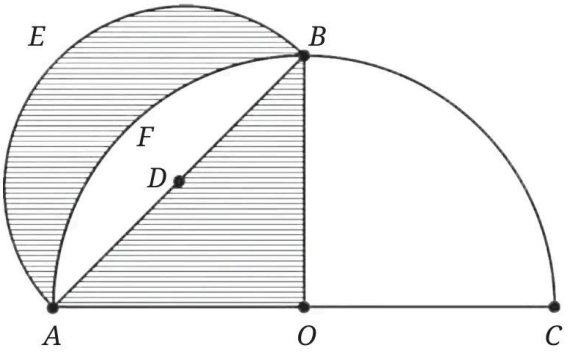


Fig. 6.55

Let OA = OB = r, so AB = r√2 and △AOB has area **½r²**.

Semicircle on AB (centre D, radius r√2/2): area = ½π(r²/2) = πr²/4.

Segment AFB (between arc AFB and chord AB) = quarter circle − △AOB = πr²/4 − ½r².

Moon AEBF = semicircle − segment = πr²/4 − (πr²/4 − ½r²) = **½r²** = area of △AOB. ∎

This "lune" was first found by Hippocrates of Chios: a curved shape with exactly the same area as a straight-sided triangle.

Recap

Formula sheet

Shape	Perimeter / length	Area
Square, side a	4a	a^2
Rectangle a × b	2(a + b)	ab
Parallelogram	2(sum of sides)	base × height
Triangle	a + b + c	½ × base × height; √[s(s−a)(s−b)(s−c)]
Trapezium	sum of sides	½(a + b)h
Rhombus / kite	sum of sides	½ × d_1 × d_2
Cyclic 4-gon	a + b + c + d	√[(s−a)(s−b)(s−c)(s−d)]
Circle, radius r	2πr = πd	πr^2
Semicircle	πr (arc) + 2r	½πr^2
Sector, angle θ	arc 2πr × θ/360, plus 2r	πr^2 × θ/360
Segment		sector − triangle

- **π** = circumference ÷ diameter, the same for every circle. π ≈ 22/7 ≈ 3.14, and π is **irrational**.
- Archimedes: $3\frac{10}{71} < \pi < 3\frac{1}{7}$. Zu Chongzhi: 355/113. Āryabhaṭa: 3.1416. Mādhava: π = 4(1 − 1/3 + 1/5 − 1/7 + …).
- A **median** divides a triangle into two triangles of equal area.
- Triangles on the same base and between the same parallels have equal area.
- If all lengths are multiplied by k, perimeter is multiplied by k and area by $\mathbf{k^2}$.

Quiz

Test yourself

Tap an option to see the right answer and the reason.

1. For every circle, circumference ÷ diameter equals…

- 2
- **π**
- 2π
- it depends on the size

C/D = π for all circles, so C = πd = 2πr.

2. The circumference of a circle of radius 14 cm (π = 22/7) is…

- 44 cm
- **88 cm**
- 616 cm
- 154 cm

2 × 22/7 × 14 = 88 cm. (616 cm^2 would be its area.)

3. An arc of a circle of radius 21 cm subtends 60°. Its length is…

- **22 cm**
- 44 cm
- 11 cm
- 66 cm

2 × 22/7 × 21 × 60/360 = 132 ÷ 6 = 22 cm.

4. Who gave the first exact formula for π, as an infinite series?

- Archimedes
- Āryabhaṭa
- **Mādhava**
- Euler

π/4 = 1 − 1/3 + 1/5 − 1/7 + … (Mādhava of Sangamagrāma).

5. Which statement is correct?

- $\pi = 22/7$
- $\pi = 3.14$
- **$\pi \approx 22/7$ and π is irrational**
- π is rational

22/7 and 3.14 are only approximations; π cannot be written as a fraction.

6. The area of a triangle with sides 5, 12, 13 cm is...

- 60 cm^2
- **30 cm^2**
- 65 cm^2
- 78 cm^2

$s = 15$, $\sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900} = 30$. (It is right-angled: $\frac{1}{2} \times 5 \times 12$.)

7. A median divides a triangle into two triangles that are always...

- congruent
- **equal in area**
- right-angled
- isosceles

Equal bases and the same height, so equal area, though usually different shapes.

8. Area of a sector of radius 14 cm and angle 90° ($\pi = 22/7$)...

- 44 cm^2
- **154 cm^2**
- 616 cm^2
- 308 cm^2

$\frac{1}{4} \times 22/7 \times 196 = 154\ cm^2$.

9. Brahmagupta's formula gives the area of...

- any quadrilateral
- **a cyclic quadrilateral**
- only a square
- a circle

√[(s−a)(s−b)(s−c)(s−d)] works only when all four vertices lie on a circle.

10. If the radius of a circle is doubled, its area becomes...

- 2 times
- 3 times
- **4 times**
- 8 times

$\pi(2r)^2 = 4\pi r^2$. Area grows with the square of the scale.

11. Circles in a row inside a rectangle (touching each other and the sides) cover what fraction of it?

- 1/2
- 3/4
- **π/4**
- depends on the number of circles

$n\pi r^2 \div (2nr \times 2r) = \pi/4 \approx 78.5\%$, for any n.

12. In Baudhāyana's construction, which identity is drawn as a picture?

- $(a + b)^2 = a^2 + 2ab + b^2$
- **$((a+b)/2)^2 - ((a-b)/2)^2 = ab$**
- $a^2 + b^2 = c^2$
- $(a - b)^2 = a^2 - 2ab + b^2$

$HP^2 = HK^2 - BH^2 = ((a+b)/2)^2 - ((a-b)/2)^2 = ab$, so the square equals the rectangle.

Notes based on NCERT Ganita Manjari (Grade 9 Mathematics, Part I), Chapter 6: Measuring Space: Perimeter and Area.
Diagrams are taken from the NCERT textbook chapter.